FIRE RESISTANT Coveralls

GUIDELINE

Fire-resistant (FR) coveralls that meet the Canadian Federal Occupational Health and Safety Code and regulations are required to conform to standards that ensure workers are adequately protected from thermal hazards, such as flames, heat, and arc flashes. The key standards for fire-resistant materials in Canada include CAN/CGSB 155.20-2017 and NFPA 2112 for protection against flash fire, as well as CSA Z462 for electrical arc flash protection.

Key Requirements for Fire-Resistant (FR) Coveralls:

1. Material Composition

FR coveralls are typically made from materials that are inherently flame-resistant or are chemically treated to provide flame-resistant properties. Common materials used include:

- Inherently Flame-Resistant Fabrics:
  - Nomex®: A meta-aramid material that is inherently flame-resistant and does not melt or drip. It maintains its FR properties for the lifetime of the garment.
  - Kevlar®: A para-aramid material often blended with Nomex® for additional strength and protection.
  - Modacrylic: A synthetic copolymer that is inherently flame-resistant and often blended with other fibers.
  - Tecasafe®: A blend of fibers that offers inherent flame resistance, including protection from electric arc and flash fire.

- Treated Cotton or Blends:
  - FR-treated cotton: Cotton or cotton blends that are chemically treated to resist ignition. Common treatments include Proban® and Pyrovatex®. These treatments can wear off over time or with laundering, requiring reapplication or replacement after a specific period.
  - FR cotton/nylon blends: Often used to balance comfort, durability, and protection. These blends combine the natural feel of cotton with the durability of nylon, treated to be flame-resistant.

2. Flame Resistance Standards

FR coveralls must be certified to specific flame resistance standards to meet Canadian regulatory requirements:

- CAN/CGSB 155.20-2017: This is the Canadian General Standards Board standard for workwear for protection against flash fire. It specifies requirements for the design, performance, and testing of garments.
  - Coveralls must provide a level of protection that allows no more than 50% body burn in a standardized test environment (such as the ASTM F1930 mannequin flash fire test).

- NFPA 2112: The North American standard for flame-resistant garments for protection against flash fire. It requires the fabric to:
  - Have a heat resistance of at least 500°F (260°C).
  - Provide less than 6% shrinkage when subjected to high temperatures.

- Achieve less than 50% total body burn in a standardized flash fire test.

- CSA Z462: For electrical arc flash protection, this standard specifies that the fabric must have a minimum arc rating (ATPV or EBT) sufficient to prevent second-degree burns when exposed to arc flash.

3. Performance Criteria

The performance of FR coveralls is evaluated based on several factors:

- Flame Resistance: The material must resist ignition, self-extinguish once the heat source is removed, and not melt or drip.
- Thermal Protection: The material should protect the wearer from thermal hazards such as flash fire, arc flash, or radiant heat.
- Durability: The FR properties of the fabric should be durable enough to withstand repeated laundering and wear.

For flash fire protection, CAN/CGSB 155.20 and NFPA 2112 standards require a Thermal Protective Performance (TPP) rating of at least 3 cal/cm², meaning the coverall must offer a level of thermal insulation to prevent burns in flash fire exposures.

4. Durability and Laundering

- Durability: Flame resistance should last for the garment’s lifetime in the case of inherently flame-resistant fabrics (e.g., Nomex®).
- Laundering: For chemically-treated fabrics, flame resistance may diminish after repeated laundering. Therefore, manufacturers specify care instructions, and the garment must be replaced once its FR properties are compromised.
- Industrial Laundering: Garments may need to be cleaned using industrial processes to maintain their flame resistance and to ensure oils, greases, and flammable contaminants do not accumulate on the fabric.

5. Marking FR Coveralls "Out of Service"

Workers must inspect FR coveralls regularly, particularly before use, to identify any damage that could compromise the protection offered by the garment. The coverall must be taken out of service if any of the following conditions are found:

- Visible damage: Cuts, tears, holes, or burns.
- Chemical contamination: Oils, fuels, or chemicals that could ignite or compromise the flame resistance.
- Loss of flame resistance: After the maximum recommended laundering cycles for FR-treated fabrics.

Once the coveralls are marked as "out of service," the worker must obtain a replacement garment before work can continue.

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Summary of Key Requirements:

1. Materials: Inherently flame-resistant (e.g., Nomex®, Tecasafe®) or FR-treated cotton/cotton blends.
2. Standards: Must comply with CAN/CGSB 155.20-2017, NFPA 2112, and CSA Z462 for specific thermal hazards.
3. Flame Resistance: Must resist ignition, self-extinguish, and not melt or drip.
4. Durability: Must maintain flame resistance over the garment's expected lifespan, with laundering requirements and periodic inspection.
5. Out of Service: Workers must inspect and replace damaged or contaminated coveralls before resuming work.

These guidelines ensure compliance with the Canadian Federal Occupational Health and Safety Code and protect workers in hazardous environments.